The minus sign indicates that $\mathbf{u}_e$ is in the opposite direction of $\mathbf{E}$.

(e)

$$N_e = -\frac{\rho_{ve}}{e} = \frac{1.81\times10^{10}}{1.6\times10^{-19}} = 1.13\times10^{29}\ \text{electrons/m}^3.$$

Exercise 4-12: Determine the density of free electrons in aluminum given that its conductivity is 3.5×10^7 (S/m) and its electron mobility is 0.0015 ($\text{m}^2/\text{V}\cdot\text{s}$).

Answer: $N_e = 1.46\times10^{29}$ electrons/m^3. (See EM.)

Exercise 4-13: The current flowing through a 100-m long conducting wire of uniform cross section has a density of 3×10^5 (A/m^2). Find the voltage drop along the length of the wire if the wire material has a conductivity of 2×10^7 (S/m).

Answer: $V = 1.5$ V. (See EM.)

4-6.2 Resistance

To demonstrate the utility of the point form of Ohm's law, we apply it to derive an expression for the resistance R of a conductor of length l and uniform cross section A, as shown in Fig. 4-16. The conductor axis is along the x direction and extends between points x_1 and x_2, with $l = x_2 - x_1$. A voltage V applied across the conductor terminals establishes an electric field $\mathbf{E} = \hat{\mathbf{x}}E_x$; the direction of $\mathbf{E}$ is from the point with higher potential (point 1 in Fig. 4-16) to the point with lower potential (point 2). The relation between V and E_x is obtained by applying Eq. (4.39):

$$V = V_1 - V_2 = -\int_{x_2}^{x_1}\mathbf{E}\cdot d\mathbf{l} = -\int_{x_2}^{x_1}\hat{\mathbf{x}}E_x\cdot\hat{\mathbf{x}}\,dl = E_x l \quad \text{(V)}. \tag{4.78}$$

Using Eq. (4.73), the current flowing through the cross section A at x_2 is

$$I = \int_A \mathbf{J}\cdot d\mathbf{s} = \int_A \sigma\mathbf{E}\cdot d\mathbf{s} = \sigma E_x A \quad \text{(A)}. \tag{4.79}$$

From $R = V/I$, the ratio of Eq. (4.78) to Eq. (4.79) gives

$$R = \frac{l}{\sigma A} \quad (\Omega). \tag{4.80}$$

We now generalize our result for R to any resistor of arbitrary shape by noting that the voltage V across the resistor is equal to the line integral of $\mathbf{E}$ over a path l between two specified points and the current I is equal to the flux of $\mathbf{J}$ through the surface S of the resistor. Thus,

$$R = \frac{V}{I} = \frac{-\int_l \mathbf{E}\cdot d\mathbf{l}}{\int_S \mathbf{J}\cdot d\mathbf{s}} = \frac{-\int_l \mathbf{E}\cdot d\mathbf{l}}{\int_S \sigma\mathbf{E}\cdot d\mathbf{s}}. \tag{4.81}$$

The reciprocal of R is called the ***conductance*** G, and the unit of G is (Ω^{-1}) or siemens (S). For the linear resistor,

$$G = \frac{1}{R} = \frac{\sigma A}{l} \quad \text{(S)}. \tag{4.82}$$

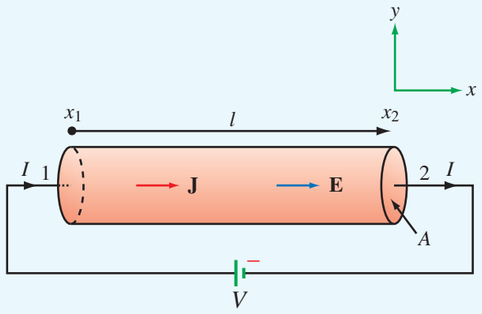


Figure 4-16 Linear resistor of cross section A and length l connected to a dc voltage source V.

Example 4-11: Conductance of Coaxial Cable

The radii of the inner and outer conductors of a coaxial cable of length l are a and b, respectively (Fig. 4-17). The insulation material has conductivity σ. Obtain an expression for G', which is the conductance per unit length of the insulation layer.

Solution: Let I be the total current flowing radially (along $\hat{\mathbf{r}}$) from the inner conductor to the outer conductor through the insulation material. At any radial distance r from the axis of the center conductor, the area through which the current flows is $A = 2\pi rl$. Hence,

$$\mathbf{J} = \hat{\mathbf{r}}\frac{I}{A} = \hat{\mathbf{r}}\frac{I}{2\pi rl}, \tag{4.83}$$

and from $\mathbf{J} = \sigma\mathbf{E}$,

$$\mathbf{E} = \hat{\mathbf{r}}\frac{I}{2\pi\sigma rl}. \tag{4.84}$$

负号表示u_e与E的方向相反。

(e)

$$N_e = -\frac{\rho_{ve}}{e} = \frac{1.81\times10^{10}}{1.6\times10^{-19}} = 1.13\times10^{29}\ \text{electrons/m}^3.$$

练习 4-12：假设铝的电导率为 3.5×10^7 (S/m)，电子迁移率为 0.0015 ($\text{m}^2/\text{V}\cdot\text{s}$)，确定铝中自由电子的密度。

Answer: $N_e = 1.46\times10^{29}$ electrons/m^3. (See EM.)

练习4-13：流过100米长、截面均匀的导线的电流密度为3×10^5 (A/m^2)。如果导线材料的电导率为 2×10^7 (S/m)，则求沿导线长度的电压降。

Answer: $V = 1.5$ V. (See EM.)

4-6.2 电阻

为了证明欧姆定律点形式的实用性，我们应用它推导了长度为 l、均匀横截面为 A 的导体的电阻 R 的表达式，如图 4-16 所示。导体轴沿 x 方向并在点 x_1 和 x_2 之间延伸，其中 $l = x_2 - x_1$。施加在导体端子上的电压 V 建立电场 $E = \hat{x}E_x$； E的方向是从电势较高的点（图4-16中的点1）到电势较低的点

潜力（第 2 点）。 V 和 E_x 之间的关系通过应用式（1）获得。 (4.39)：

$$V = V_1 - V_2 = -\int_{x_2}^{x_1}\mathbf{E}\cdot d\mathbf{l} = -\int_{x_2}^{x_1}\hat{\mathbf{x}}E_x\cdot\hat{\mathbf{x}}\,dl = E_x l \quad \text{(V)}. \tag{4.78}$$

使用方程式（4.73），流过 x_2 处横截面 A 的电流为

$$I = \int_A \mathbf{J}\cdot d\mathbf{s} = \int_A \sigma\mathbf{E}\cdot d\mathbf{s} = \sigma E_x A \quad \text{(A)}. \tag{4.79}$$

从 R = V /I 可以得出方程式的比率： (4.78) 至方程式(4.79) 给出

$$R = \frac{l}{\sigma A} \quad (\Omega). \tag{4.80}$$

现在，我们将 R 的结果推广到任意形状的电阻器，注意电阻器两端的电压 V 等于 E 在两个指定点之间的路径 l 上的线积分，电流 I 等于通过电阻器表面 S 的通量 J。因此，

$$R = \frac{V}{I} = \frac{-\int_l \mathbf{E}\cdot d\mathbf{l}}{\int_S \mathbf{J}\cdot d\mathbf{s}} = \frac{-\int_l \mathbf{E}\cdot d\mathbf{l}}{\int_S \sigma\mathbf{E}\cdot d\mathbf{s}}. \tag{4.81}$$

R的倒数称为电导G，G的单位为(Ω^{-1})或西门子(S)。对于线性电阻，

$$G = \frac{1}{R} = \frac{\sigma A}{l} \quad \text{(S)}. \tag{4.82}$$

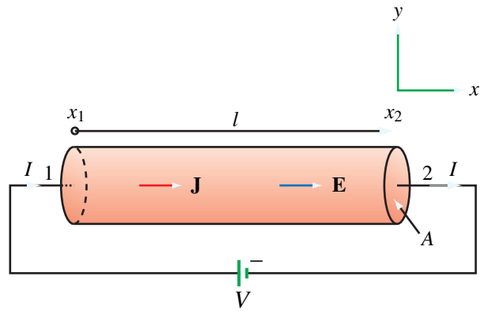


图 4-16 连接到直流电压源 V 的横截面为 A、长度为 l 的线性电阻。

例 4-11：同轴电缆的电导

长度为l的同轴电缆的内导体和外导体的半径分别为a和b（图4-17）。绝缘材料具有导电率σ。求出G′的表达式，即绝缘层单位长度的电导。

解：设 I 为从内导体通过绝缘材料径向（沿 r̂）流到外导体的总电流。在距中心导体轴任意径向距离 r 处，电流流过的面积为 A = 2π rl。因此，

$$\mathbf{J} = \hat{\mathbf{r}}\frac{I}{A} = \hat{\mathbf{r}}\frac{I}{2\pi rl}, \tag{4.83}$$

并从J = σ E，

$$\mathbf{E} = \hat{\mathbf{r}}\frac{I}{2\pi\sigma rl}. \tag{4.84}$$

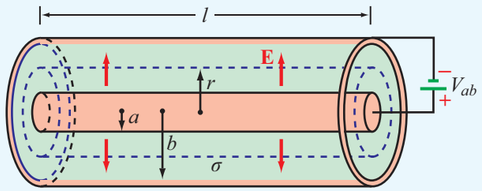


Figure 4-17 Coaxial cable of Example 4-11.

In a resistor, the current flows from higher electric potential to lower potential. Hence, if **J** is in the $\hat{\mathbf{r}}$ direction, the inner conductor must be at a potential higher than that at the outer conductor. Accordingly, the voltage difference between the conductors is

$$V_{ab} = -\int_b^a \mathbf{E}\cdot d\mathbf{l} = -\int_b^a \frac{I}{2\pi\sigma l}\frac{\hat{\mathbf{r}}\cdot\hat{\mathbf{r}}\,dr}{r} = \frac{I}{2\pi\sigma l}\ln\left(\frac{b}{a}\right). \tag{4.85}$$

The conductance per unit length is then

$$G' = \frac{G}{l} = \frac{1}{Rl} = \frac{I}{V_{ab}l} = \frac{2\pi\sigma}{\ln(b/a)} \quad \text{(S/m)}. \tag{4.86}$$

4-6.3 Joule's Law

We now consider the power dissipated in a conducting medium in the presence of an electrostatic field **E**. The medium contains free electrons and holes with volume charge densities ρ_{ve} and ρ_{vh}, respectively. The electron and hole charge contained in an elemental volume $\Delta\upsilon$ is $q_e = \rho_{ve}\,\Delta\upsilon$ and $q_h = \rho_{vh}\,\Delta\upsilon$, respectively. The electric forces acting on q_e and q_h are $\mathbf{F}_e = q_e\mathbf{E} = \rho_{ve}\mathbf{E}\,\Delta\upsilon$ and $\mathbf{F}_h = q_h\mathbf{E} = \rho_{vh}\mathbf{E}\,\Delta\upsilon$. The work (energy) expended by the electric field in moving q_e a differential distance Δl_e and moving q_h a distance Δl_h is

$$\Delta W = \mathbf{F}_e\cdot\Delta\mathbf{l}_e + \mathbf{F}_h\cdot\Delta\mathbf{l}_h. \tag{4.87}$$

Power P is measured in watts (W) and is defined as the time rate of change of energy. The power corresponding to ΔW is

$$\begin{aligned}\Delta P = \frac{\Delta W}{\Delta t} &= \mathbf{F}_e\cdot\frac{\Delta\mathbf{l}_e}{\Delta t} + \mathbf{F}_h\cdot\frac{\Delta\mathbf{l}_h}{\Delta t}\\ &= \mathbf{F}_e\cdot\mathbf{u}_e + \mathbf{F}_h\cdot\mathbf{u}_h\\ &= (\rho_{ve}\mathbf{E}\cdot\mathbf{u}_e + \rho_{vh}\mathbf{E}\cdot\mathbf{u}_h)\,\Delta\upsilon = \mathbf{E}\cdot\mathbf{J}\,\Delta\upsilon,\end{aligned} \tag{4.88}$$

where $\mathbf{u}_e = \Delta\mathbf{l}_e/\Delta t$ and $\mathbf{u}_h = \Delta\mathbf{l}_h/\Delta t$ are the electron and hole drift velocities, respectively. Equation (4.75) was used in the last step of the derivation leading to Eq. (4.88). For a volume υ, the total dissipated power is

$$P = \int_\upsilon \mathbf{E}\cdot\mathbf{J}\,d\upsilon \quad \text{(W)} \quad \textbf{(Joule's law)}, \tag{4.89}$$

and in view of Eq. (4.73),

$$P = \int_\upsilon \sigma|\mathbf{E}|^2\,d\upsilon \quad \text{(W)}. \tag{4.90}$$

Equation (4.89) is a mathematical statement of ***Joule's law***. For the resistor example considered earlier, $|\mathbf{E}| = E_x$ and its volume is $\upsilon = lA$. Separating the volume integral in Eq. (4.90) into a product of a surface integral over A and a line integral over l, we have

$$\begin{aligned}P = \int_\upsilon \sigma|\mathbf{E}|^2\,d\upsilon &= \int_A \sigma E_x\,ds\int_l E_x\,dl\\ &= (\sigma E_x A)(E_x l) = IV \quad \text{(W)},\end{aligned} \tag{4.91}$$

where use was made of Eq. (4.78) for the voltage V and Eq. (4.79) for the current I. With $V = IR$, we obtain the familiar expression

$$P = I^2R \quad \text{(W)}. \tag{4.92}$$

Concept Question 4-17: What is the fundamental difference between an insulator, a semiconductor, and a conductor?

Concept Question 4-18: Show that the power dissipated in the coaxial cable of **Fig. 4-17** is

$$P = \frac{I^2\ln(b/a)}{2\pi\sigma l}.$$

Exercise 4-14: A 50-m long copper wire has a circular cross section with radius $r = 2$ cm. Given that the conductivity of copper is 5.8×10^7 S/m, determine (a) the resistance R of the wire and (b) the power dissipated in the wire if the voltage across its length is 1.5 mV.

Answer: (a) $R = 6.9\times10^{-4}\ \Omega$, (b) $P = 3.3$ mW. (See EM.)

Exercise 4-15: Repeat part (b) of Exercise 4-14 by applying Eq. (4.90). (See EM.)

4-7 Dielectrics

The fundamental difference between a conductor and a dielectric is that electrons in the outermost atomic shells of

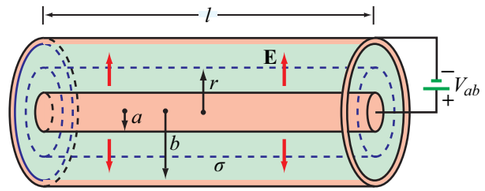


图4-17 例4-11的同轴电缆。

在电阻器中，电流从较高电势流向较低电势。因此，如果 J 沿 r̂ 方向，则内导体的电势必定高于外导体的电势。因此，导体之间的电压差为

$$V_{ab} = -\int_b^a \mathbf{E}\cdot d\mathbf{l} = -\int_b^a \frac{I}{2\pi\sigma l}\frac{\hat{\mathbf{r}}\cdot\hat{\mathbf{r}}\,dr}{r} = \frac{I}{2\pi\sigma l}\ln\left(\frac{b}{a}\right). \tag{4.85}$$

那么单位长度的电导为

$$G' = \frac{G}{l} = \frac{1}{Rl} = \frac{I}{V_{ab}l} = \frac{2\pi\sigma}{\ln(b/a)} \quad \text{(S/m)}. \tag{4.86}$$

4-6.3 焦耳定律

现在我们考虑存在静电场 E 时导电介质中耗散的功率。该介质包含体积电荷密度分别为 ρ_{ve} 和 ρ_{vh} 的自由电子和空穴。元素体积 $\Delta\upsilon$ 中包含的电子和空穴电荷分别为 $q_e = \rho_{ve}\,\Delta\upsilon$ 和 $q_h = \rho_{vh}\,\Delta\upsilon$。作用在 q_e 和 q_h 上的电力为 $F_e = q_eE = \rho_{ve}E\,\Delta\upsilon$ 和 $F_h = q_hE = \rho_{vh}E\,\Delta\upsilon$。电场将 q_e 移动微分距离 Δl_e 和将 q_h 移动距离 Δl_h 所消耗的功（能量）为

$$\Delta W = \mathbf{F}_e\cdot\Delta\mathbf{l}_e + \mathbf{F}_h\cdot\Delta\mathbf{l}_h. \tag{4.87}$$

功率 P 以瓦 (W) 为单位测量，定义为能量的时间变化率。ΔW对应的功率为

$$\begin{aligned}\Delta P = \frac{\Delta W}{\Delta t} &= \mathbf{F}_e\cdot\frac{\Delta\mathbf{l}_e}{\Delta t} + \mathbf{F}_h\cdot\frac{\Delta\mathbf{l}_h}{\Delta t}\\ &= \mathbf{F}_e\cdot\mathbf{u}_e + \mathbf{F}_h\cdot\mathbf{u}_h\\ &= (\rho_{ve}\mathbf{E}\cdot\mathbf{u}_e + \rho_{vh}\mathbf{E}\cdot\mathbf{u}_h)\,\Delta\upsilon = \mathbf{E}\cdot\mathbf{J}\,\Delta\upsilon,\end{aligned} \tag{4.88}$$

其中 $u_e = \Delta l_e/\Delta t$ 和 $u_h = \Delta l_h/\Delta t$ 分别是电子和空穴漂移速度。使用方程（4.75）

在导出方程的最后一步中（4.88）。对于体积 υ，总耗散功率为

$$P = \int_\upsilon \mathbf{E}\cdot\mathbf{J}\,d\upsilon \quad \text{(W)} \quad \textbf{(Joule's law)}, \tag{4.89}$$

并考虑到等式。(4.73),

$$P = \int_\upsilon \sigma|\mathbf{E}|^2\,d\upsilon \quad \text{(W)}. \tag{4.90}$$

方程（4.89）是焦耳定律的数学表述。对于前面考虑的电阻器示例，为 |E| = E_x，其体积为 υ = lA。分离方程中的体积积分。将 (4.90) 代入 A 上的面积分和 l 上的线积分的乘积，我们有

$$\begin{aligned}P = \int_\upsilon \sigma|\mathbf{E}|^2\,d\upsilon &= \int_A \sigma E_x\,ds\int_l E_x\,dl\\ &= (\sigma E_x A)(E_x l) = IV \quad \text{(W)},\end{aligned} \tag{4.91}$$

其中使用了方程式。(4.78) 为电压 V 和方程。(4.79) 为电流 I。通过 V = IR，我们得到熟悉的表达式

$$P = I^2R \quad \text{(W)}. \tag{4.92}$$

概念问题 4-17：绝缘体、半导体和导体之间的根本区别是什么？

概念问题 4-18：证明图 4-17 的同轴电缆中消耗的功率为

$$P = \frac{I^2\ln(b/a)}{2\pi\sigma l}.$$

练习 4-14：一根 50 米长的铜线，其横截面为圆形，半径 r = 2 厘米。假定铜的电导率为 5.8×10^7 S/m，请确定 (a) 导线的电阻 R 和 (b) 如果导线长度上的电压为 1.5 mV，则导线中消耗的功率。

Answer: (a) $R = 6.9\times10^{-4}\ \Omega$, (b) $P = 3.3$ mW. (See EM.)

练习 4-15：通过应用方程 4 重复练习 4-14 的 (b) 部分。（4.90）。（参见EM。）

4-7 电介质

导体和电介质之间的根本区别在于，电子位于导体的最外层原子壳层中。

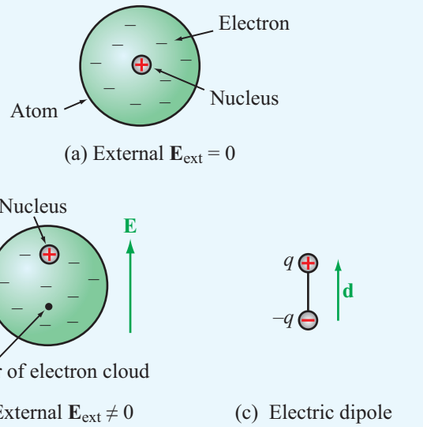


Figure 4-18 In the absence of an external electric field **E**, the center of the electron cloud is co-located with the center of the nucleus, but when a field is applied, the two centers are separated by a distance *d*.

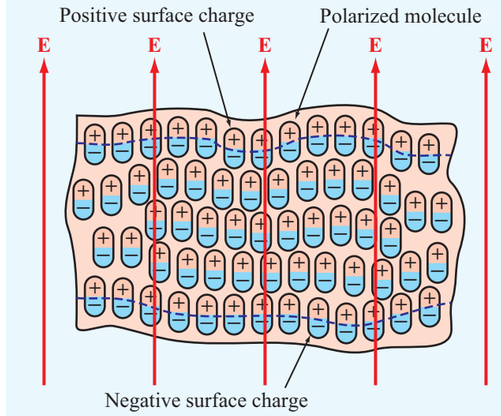


Figure 4-19 A dielectric medium polarized by an external electric field **E**.

a conductor are only weakly tied to atoms and hence can freely migrate through the material, whereas in a dielectric they are strongly bound to the atom. In the absence of an electric field, the electrons in ***nonpolar*** molecules form a symmetrical cloud around the nucleus, with the center of the cloud coinciding with the nucleus (**Fig. 4-18(a)**). The electric field generated by the positively charged nucleus attracts and holds the electron cloud around it, and the mutual repulsion of the electron clouds of adjacent atoms shapes its form. When a conductor is subjected to an externally applied electric field, the most loosely bound electrons in each atom can jump from one atom to the next, thereby setting up an electric current. In a dielectric, however, an externally applied electric field **E** cannot effect mass migration of charges since none are able to move freely. Instead, **E** will ***polarize*** the atoms or molecules in the material by moving the center of the electron cloud away from the nucleus (**Fig. 4-18(b)**). The polarized atom or molecule may be represented by an electric dipole consisting of charges $+q$ in the nucleus and $-q$ at the center of the electron cloud (**Fig. 4-18(c)**). Each such dipole sets up a small electric field pointing from the positively charged nucleus to the center of the equally but negatively charged electron cloud. This ***induced*** electric field, called a ***polarization*** field, generally is weaker than and opposite in direction to **E**. Consequently, the net electric field present in the dielectric material is smaller than **E**. At the microscopic level, each dipole exhibits a dipole moment similar to that described in Example 4-9. Within a block of dielectric material subject to a uniform external field, the dipoles align themselves linearly, as shown in **Fig. 4-19**. Along the upper and lower edges of the material, the dipole arrangement exhibits positive and negative surface charge densities, respectively.

It is important to stress that this description applies to only nonpolar molecules that do not have permanent dipole moments. Nonpolar molecules become polarized only when an external electric field is applied; when the field is removed, the molecules return to their original unpolarized state.

In polar materials such as water, the molecules possess built-in ***permanent dipole moments*** that are randomly oriented in the absence of an applied electric field, and owing to their random orientations, the dipoles of polar materials produce no net macroscopic dipole moment (at the macroscopic scale, each point in the material represents a small volume containing thousands of molecules). Under the influence of an applied field, the permanent dipoles tend to align themselves along the direction of the electric field in a manner similar to that shown in **Fig. 4-19** for nonpolar materials.

4-7.1 Polarization Field

In free space $\mathbf{D} = \varepsilon_0\mathbf{E}$, the presence of microscopic dipoles in a dielectric material alters that relationship to

$$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}, \tag{4.93}$$

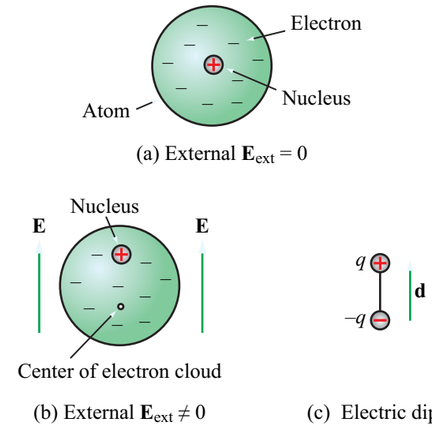


图4-18 在没有外部电场E的情况下，电子云的中心与原子核的中心位于同一位置，但是当施加电场时，两个中心相距距离d。

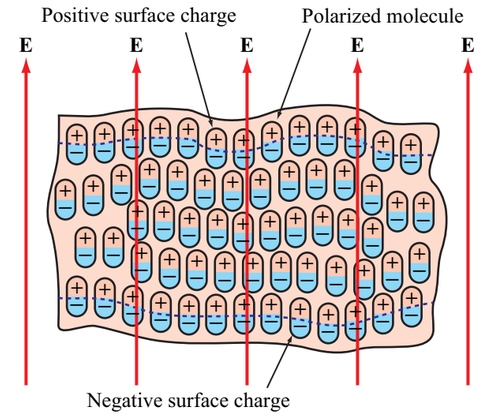


图 4-19 被外部电场 E 极化的介电介质

导体仅与原子弱结合，因此可以在材料中自由迁移，而在电介质中，它们与原子强结合。在没有电场的情况下，非极性分子中的电子在原子核周围形成对称的云，云的中心与原子核重合（图4-18（a））。带正电的原子核产生的电场吸引并保持其周围的电子云，相邻原子的电子云的相互排斥决定了其形状。当导体受到外部施加的电场时，每个原子中束缚最松的电子可以从一个原子跳到下一个原子，从而产生电流。然而，在电介质中，外部施加的电场 E 不能影响电荷的质量迁移，因为没有电荷能够自由移动。相反，E 会通过将电子云的中心移离原子核来极化材料中的原子或分子（图 4-18(b)）。极化的原子或分子可以用电偶极子来表示，该电偶极子由原子核中的电荷+q和电子云中心的−q组成（图4-18（c））。每个这样的偶极子都会建立一个小电场，从带正电的原子核指向等量但带负电的电子云的中心。这种感应电场称为极化场，通常比 E 弱且方向相反。因此，介电材料中存在的净电场小于 E。在微观层面上，每个偶极子表现出类似于实施例4-9中所述的偶极矩。在一块受到均匀外场作用的介电材料内，偶极子会自行线性排列，如图 4-19 所示。沿着材料的上边缘和下边缘，偶极子排列分别表现出正表面电荷密度和负表面电荷密度。

需要强调的是，这种描述仅适用于不具有永久偶极矩的非极性分子。非极性分子只有在施加外部电场时才会极化；当电场移除时，分子恢复到原来的非极化状态。

在水等极性材料中，分子具有内置的永久偶极矩，在没有施加电场的情况下，这些偶极矩会随机定向，并且由于其随机定向，极性材料的偶极子不会产生净宏观偶极矩（在宏观尺度上，材料中的每个点代表包含数千个分子的小体积）。在外加电场的影响下，永久偶极子倾向于沿着电场方向排列，类似于图 4-19 所示的非极性材料。

4-7.1 偏振场

在自由空间 $D = \varepsilon_0 E$ 中，介电材料中微观偶极子的存在改变了这种关系：

$$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}, \tag{4.93}$$

where **P**, called the ***electric polarization field***, accounts for the polarization properties of the material. The polarization field is produced by the electric field **E** and depends on the material properties. A dielectric medium is said to be ***linear*** if the magnitude of the induced polarization field **P** is directly proportional to the magnitude of **E**, and ***isotropic*** if **P** and **E** are in the same direction. Some crystals allow more polarization to take place along certain directions, such as the crystal axes, than along others. In such ***anisotropic*** dielectrics, **E** and **P** may have different directions. A medium is said to be ***homogeneous*** if its constitutive parameters (ε, μ, and σ) are constant throughout the medium. Our present treatment will be limited to media that are linear, isotropic, and homogeneous. For such media, **P** is directly proportional to **E** and is expressed as

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}, \tag{4.94}$$

where χ_e is called the ***electric susceptibility*** of the material. Inserting Eq. (4.94) into Eq. (4.93), we have

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \varepsilon_0 \chi_e \mathbf{E} = \varepsilon_0 (1 + \chi_e)\mathbf{E} = \varepsilon \mathbf{E}, \tag{4.95}$$

which defines the permittivity ε of the material as

$$\varepsilon = \varepsilon_0 (1 + \chi_e). \tag{4.96}$$

It is often convenient to characterize the permittivity of a material relative to that of free space, ε_0; this is accommodated by the relative permittivity $\varepsilon_r = \varepsilon / \varepsilon_0$. Values of ε_r are listed in **Table 4-2** for a few common materials, and a longer list is given in Appendix B. In free space $\varepsilon_r = 1$, and *for most conductors*, $\varepsilon_r \approx 1$. The dielectric constant of air is approximately 1.0006 at sea level and decreases toward unity with increasing altitude. Except in some special circumstances, such as when calculating electromagnetic wave refraction (bending) through the atmosphere over long distances, *air can be treated as if it were free space.*

4-7.2 Dielectric Breakdown

The preceding dielectric–polarization model presumes that the magnitude of **E** does not exceed a certain critical value, which is known as the ***dielectric strength*** E_{ds} of the material. Beyond this, electrons will detach from the molecules and accelerate through the material in the form of a conduction current. When this happens, sparking can occur, and the dielectric material can sustain permanent damage due to electron collisions with the molecular structure. This abrupt change in behavior is called ***dielectric breakdown***.

▶ The dielectric strength E_{ds} is the largest magnitude of **E** that the material can sustain without breakdown. ◀

Dielectric breakdown can occur in gases, liquids, and solids. The dielectric strength E_{ds} depends on the material composition, as well as other factors such as temperature and humidity. For air, E_{ds} is roughly 3 (MV/m); for glass, 25 to 40 (MV/m); and for mica, 200 (MV/m) (see **Table 4-2**).

A charged thundercloud at electric potential V relative to the ground induces an electric field $E = V/d$ in the air beneath it, where d is the height of the cloud base above the ground. If V is sufficiently large so that E exceeds the dielectric strength of air, ionization occurs and a lightning discharge follows. The ***breakdown voltage*** V_{br} of a parallel-plate capacitor is discussed in Example 4-12.

Example 4-12: Dielectric Breakdown

In a parallel-plate capacitor with a separation d between the conducting plates, the electric field E in the dielectric material

Table 4-2 Relative permittivity (dielectric constant) and dielectric strength of common materials.

Material	Relative Permittivity, ε_r	Dielectric Strength, E_{ds} (MV/m)
Air (at sea level)	1.0006	3
Petroleum oil	2.1	12
Polystyrene	2.6	20
Glass	4.5–10	25–40
Quartz	3.8–5	30
Bakelite	5	20
Mica	5.4–6	200

Note: $\varepsilon = \varepsilon_r \varepsilon_0$ and $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.

其中 P，称为电极化场，说明了材料的极化特性。极化场由电场 E 产生，取决于材料特性。如果感应极化场 P 的大小与 E 的大小成正比，则称介电介质是线性的；如果 P 和 E 方向相同，则称介电介质是各向同性的。有些晶体允许沿着某些方向（例如晶轴）比沿着其他方向发生更多的偏振。在这种各向异性电介质中，E和P可以具有不同的方向。如果介质的本构参数（ε、μ 和 σ）在整个介质中恒定，则称该介质是均匀的。我们目前的处理仅限于线性、各向同性和均匀的介质。对于此类介质，P 与 E 成正比，表示为

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}, \tag{4.94}$$

其中χ_e称为材料的电化率。插入方程式（4.94）代入等式。（4.93），我们有

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \varepsilon_0 \chi_e \mathbf{E} = \varepsilon_0 (1 + \chi_e)\mathbf{E} = \varepsilon \mathbf{E}, \tag{4.95}$$

将材料的介电常数 ε 定义为

$$\varepsilon = \varepsilon_0 (1 + \chi_e). \tag{4.96}$$

通常可以方便地表征材料相对于自由空间的介电常数 ε_0；这是通过相对介电常数 $\varepsilon_r = \varepsilon / \varepsilon_0$ 来实现的。表 4-2 中列出了一些常见材料的 ε_r 值，附录 B 中给出了更长的列表。在自由空间中 $\varepsilon_r = 1$，对于大多数导体，$\varepsilon_r \approx 1$。空气的介电常数在海平面处约为 1.0006，并随着高度的增加而减小到 1。除某些特殊情况外，例如计算长距离穿过大气层的电磁波折射（弯曲）时，空气可以视为自由空间。

4-7.2 介质击穿

前面的介电极化模型假设 E 的大小不超过某个临界值，该临界值称为材料的介电强度 E_{ds}。除此之外，电子将从分子中分离出来，并以传导电流的形式加速穿过材料。发生这种情况时，可能会产生火花，并且电介质材料可能会因电子与分子结构的碰撞而遭受永久性损坏。这种行为的突然变化称为介电击穿。

▶ 介电强度 E_{ds} 是材料在不击穿的情况下可以承受的最大 E 值。 ◀

气体、液体和固体中都可能发生电介质击穿。介电强度 E_{ds} 取决于材料成分以及温度和湿度等其他因素。对于空气，E_{ds} 大约为 3 (MV/m)；对于玻璃，25至40（MV/m）；云母为200（MV/m）（见表4-2）。

相对于地面的电势为 V 的带电雷云会在其下方的空气中感应出电场 $E = V/d$，其中 d 是云底距地面的高度。如果 V 足够大，使得 E 超过空气的介电强度，就会发生电离并随后发生闪电放电。示例 4-12 讨论了平行板电容器的击穿电压 V_{br}。

例 4-12： 介电击穿

在导电板之间间距为 d 的平行板电容器中，介电材料中的电场 E

表4-2 常见材料的相对介电常数（介电常数）和介电强度。

Material	Relative Permittivity, ε_r	Dielectric Strength, E_{ds} (MV/m)
Air (at sea level)	1.0006	3
Petroleum oil	2.1	12
Polystyrene	2.6	20
Glass	4.5–10	25–40
Quartz	3.8–5	30
Bakelite	5	20
Mica	5.4–6	200

注：$\varepsilon = \varepsilon_r \varepsilon_0$ 和 $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m。

separating the two plates is related to the voltage V between the two plates by

$$E = \frac{V}{d}.$$

The breakdown voltage V_{br} corresponds to the value of V at which $E = E_{\text{ds}}$, where E_{ds} is the dielectric strength of the material contained between the plates. That is,

$$V_{\text{br}} = E_{\text{ds}} d.$$

If V exceeds V_{br}, the electric charges will "spark" their way between the two plates.

A thin capacitor filled with quartz operates at 60 V. If $d = 0.01$ mm, what is the breakdown voltage, and how does it compare with the operating voltage?

Solution: From **Table 4-2**, $E_{\text{ds}} = 30 \times 10^6$ V/m for quartz. Hence, the breakdown voltage is

$$V_{\text{br}} = E_{\text{ds}} d = 30 \times 10^6 \times 10^{-5} = 300 \text{ V},$$

which is much higher than the operating voltage of 60 V. Therefore, the capacitor should experience no issues with dielectric breakdown.

Concept Question 4-19: What is a polar material? A nonpolar material?

Concept Question 4-20: Do **D** and **E** always point in the same direction? If not, when do they not?

Concept Question 4-21: What happens when dielectric breakdown occurs?

4-8 Electric Boundary Conditions

▶ A vector field is said to be ***spatially continuous*** if it does not exhibit abrupt changes in either magnitude or direction as a function of position. ◀

Even though the electric field may be continuous in adjoining dissimilar media, it may well be discontinuous at the boundary between them. Boundary conditions specify how the components of fields tangential and normal to an interface between two media relate across the interface Here we derive a general set of boundary conditions for **E**, **D**, and **J** that is applicable at the interface between *any* two dissimilar media—be they two dielectrics or a conductor and a dielectric. Of course, any of the dielectrics may be free space. Even though these boundary conditions are derived assuming electrostatic conditions, they remain valid for time-varying electric fields as well. **Figure 4-20** shows an interface between medium 1 with permittivity ε_1 and medium 2 with permittivity ε_2. In the general case, the interface may contain a surface charge density ρ_s (unrelated to the dielectric polarization charge density).

To derive the boundary conditions for the tangential components of **E** and **D**, we consider the closed rectangular loop *abcda* shown in **Fig. 4-20** and apply the conservative property of the electric field expressed by Eq. (4.40), which states that the line integral of the electrostatic field around a closed path is always zero. By letting $\Delta h \to 0$, the contributions to the line integral by segments *bc* and *da* vanish. Hence,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{E}_1 \cdot \hat{\boldsymbol{\ell}}_1 \, dl + \int_c^d \mathbf{E}_2 \cdot \hat{\boldsymbol{\ell}}_2 \, dl = 0, \qquad (4.97)$$

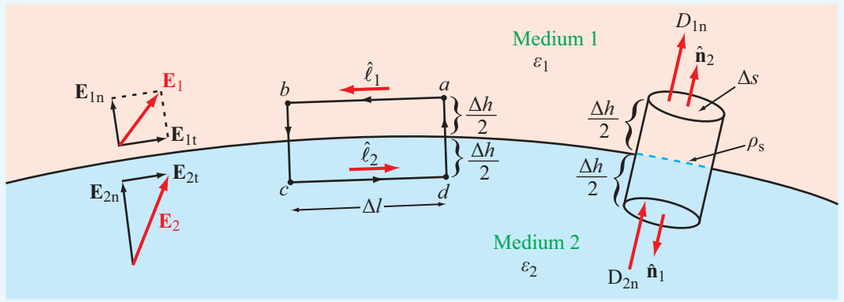


Figure 4-20 Interface between two dielectric media.

两板之间的距离与两板之间的电压 V 相关：

$$E = \frac{V}{d}.$$

击穿电压 V_{br} 对应于 $E = E_{\text{ds}}$ 时的 V 值，其中 E_{ds} 是板之间所含材料的介电强度。那是，

$$V_{\text{br}} = E_{\text{ds}} d.$$

如果 V 超过 V_{br}，电荷将在两个板之间“激发”。

填充石英的薄电容器在 60 V 下工作。如果 d = 0.01 m m，击穿电压是多少？它与工作电压相比如何？

解：根据表 4-2，石英的 $E_{\text{ds}} = 30 \times 10^6$ V/m。因此，击穿电压为

$$V_{\text{br}} = E_{\text{ds}} d = 30 \times 10^6 \times 10^{-5} = 300 \text{ V},$$

该电压远高于 60 V 的工作电压。因此，电容器不应出现介电击穿问题。

概念问题 4-19：什么是极性材料？非极性材料？

概念问题 4-20：D 和 E 总是指向同一个方向吗？如果没有，什么时候没有？

概念问题 4-21：发生电介质击穿时会发生什么？

4-8 电边界条件

▶ 如果矢量场在大小或方向上不随位置发生突变，则称该矢量场是空间连续的。 ◀

尽管电场在相邻的不同介质中可能是连续的，但在它们之间的边界处很可能是不连续的。边界条件指定与两种介质之间的界面相切和法向的场分量如何在界面上相关。这里我们推导出 E、D 和 J 的一组通用边界条件，适用于任何两种不同介质（无论是两种电介质还是导体和电介质）之间的界面。当然，任何电介质都可以是自由空间。尽管这些边界条件是在假设静电条件的情况下得出的，但它们对于时变电场仍然有效。图 4-20 显示了介电常数为 ε_1 的介质 1 和介电常数为 ε_2 的介质 2 之间的界面。在一般情况下，界面可以包含与介电极化电荷密度)无关的表面电荷密度ρ_s（。

为了导出 E 和 D 切向分量的边界条件，我们考虑图 4-20 中所示的闭矩形环 abcda，并应用方程 1 表示的电场的保守性质。（4.40），它指出闭合路径周围静电场的线积分始终为零。通过让 $\Delta h \to 0$，线段 bc 和 da 对线积分的贡献消失。因此，

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{E}_1 \cdot \hat{\boldsymbol{\ell}}_1 \, dl + \int_c^d \mathbf{E}_2 \cdot \hat{\boldsymbol{\ell}}_2 \, dl = 0, \qquad (4.97)$$

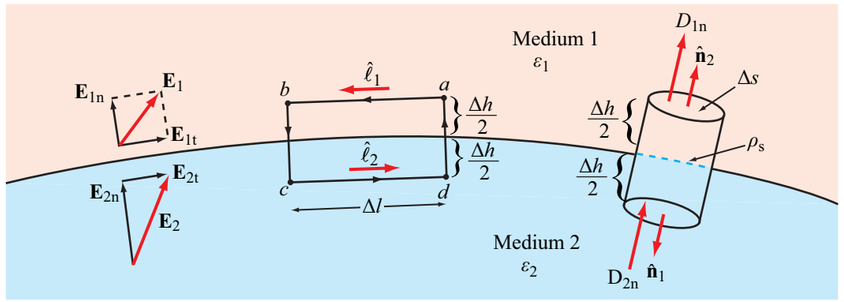


图 4-20 两种电介质之间的界面。